

Plan 4 of 4 — Track S: mass pattern search, walk-forward validated

Written 2026-08-18. Live parameters: `configs/scan.yml`. This document explains; the config defines.

0. Why this document exists

Plans 1–3 describe Track D — the institutional deal event studies. Their machinery is built: `power.py`, `split.py`, `multiplicity.py`, `design.py`, 146 tests.

None of it serves the mass-search side of the project, which the owner correctly identified on 2026-08-18 as being half the point and absent from the system. The 31.9M combinations existed in configs, two decision records and a paragraph of prose, with zero lines of code behind them.

This plan corrects that. It is the design for Track S, which runs in parallel with Track D and shares the registration, decision and multiplicity layers.

1. What Track S is, in one paragraph

Search a very large space of candidate patterns — calendar patterns and combinations of trading signals — inside a training window, then require each candidate to repeat in a window it has never seen. Do this across many folds. **Report as the headline not which patterns survived, but how well the search procedure itself performed out of sample.** Report surviving patterns second.

2. What the predecessor did, and the one thing it never asked

MICCV2 scanned **31,893,556 cells**: 13 windows × 242 calendar ordinals × 4,200 symbols, plus index-level variants across 4 alignments and 2 bases.

Its own verdict, from `docs/SEASONALITY_FINDINGS.md` :

"Essentially none of it survives contact with its own null. The single best pattern found in NIFTY50 — a 3-day window that rose in 94.7% of years — sits at the 94th percentile of what randomly rotated data produces, which is to say it is an ordinary result of looking 31.9 million times."

And the finding that matters most:

"The scan of millions found nothing. The eight pre-registered guesses found two. That asymmetry is the finding."

Both of those two were subsequently killed by the corrected fee schedule — turn-of-month goes from **+3.70 to -6.36 bps** net.

What it never asked. Every one of those 31.9M cells was scored against a single sample. The question was *"is this cell better than chance in-sample?"* It was never *"does this cell do it again?"* —

which is the only question a seasonal or signal claim actually makes.

That is the gap this track closes.

3. Three measurements that determined the design

Everything below was measured on 17–18 August 2026 before any of this plan was written. Under the project's own Rule 1, no design element enters a plan without a computed number beside it.

3.1 A calendar cell fires once a year, and that is fatal per stock

Fold	Train obs	Test obs
train 2005–10 → test 2010–12	5	2
train 2005–15 → test 2015–20	10	5
train 2005–18 → test 2018–21	13	3

Two observations detect an effect of **49.5% per year**. Five detect 31.3%. A stock's entire 21-year history detects 15.3%.

The history is not there either. Distribution of price history across 4,200 symbols:

Percentile	Years
25th	2.3
50th	5.5
75th	14.7
90th	21.0

Only **513 of 4,200** have 20+ years — and "survived twenty years" is itself a survivorship-selected sample.

Conclusion: per-stock calendar analysis is not difficult, it is arithmetically impossible. It is still run and reported, marked UNDERPOWERED with its MDE beside it, because "we could not tell" is a more useful statement than silence.

3.2 Pooling across stocks does not rescue it — and the obvious fix is undefined

The obvious response is to pool: 4,200 stocks × 5 years = 21,000 observations, MDE 0.48%/yr. That reasoning fails twice, and both failures are instructive.

First failure — raw prices. Every stock experiences trading day 47 **on the same day**, so all of them share whatever the market did. Measured mean pairwise correlation of raw daily returns across 656 liquid names, 2015+: $\rho = +0.2350$, giving n_{eff} of **4.3** out of 21,000 and an MDE of **33.95%/yr**. Raw-price pooling is dead. The predecessor treated PRICE and MARKET_RELATIVE as equal bases; its entire PRICE-basis pooled output was arithmetically worthless.

Second failure — and this one I got wrong first time. An earlier draft of this plan reported that market-relative returns have $\rho = +0.0001$, "indistinguishable from independence", and concluded that removing the market factor multiplies effective sample size by $\sim 1,550\times$.

That was an artifact. Subtracting the cross-sectional mean forces average pairwise correlation to $-1/(N-1)$ *regardless of the input*. Verified against simulated controls:

Input	Raw ρ	After demeaning
pure independence, N=657	-0.00003	-0.00148
strong market factor, N=657	+0.4035	-0.00145
theoretical $-1/(N-1)$	—	-0.00152

The statistic cannot distinguish a strong market factor from pure noise. It measured the arithmetic of subtraction.

And the real consequence is larger than the erratum. If market-relative means "minus the cross-sectional mean", then the cross-sectional mean *of* market-relative returns is identically zero. Measured across 656 stocks and 2,870 sessions, the largest absolute value on any day is **1.698 × 10⁻¹⁷**.

"Is trading day 47 good on average, market-relative?" is not a weak question. It has no content. The answer is exactly zero for every day, by construction.

So the pooled-average formulation is not underpowered — it is *undefined*. Which means the cross-sectional formulation in §5 is not the strongest option among several. **It is the only one.** See decision 0021.

3.3 Hundreds of folds are not hundreds of tests

Anchored expanding windows share almost all of their training data — consecutive folds are roughly 95% the same fit.

Design	Folds	Independent tests
anchored, 2y test, 1y step, 2010–2026	16	8
anchored, 2y test, 2y step (disjoint)	8	8
anchored, 1y test, 1y step (disjoint)	16	16
CPCV, N=16 groups, k=2	120	16
CPCV, N=20 groups, k=2	190	20

So "make some hundreds of tests" is achievable, but the honest evidence count is 8–20, not 120–190. Both designs are run because they answer different questions:

- **Sequential walk-forward** answers "*would this have worked if I had deployed it in real time?*" — the deployment question, and the one originally asked.
- **CPCV** yields many more paths, which is what a stable estimate of overfitting probability requires.

Neither substitutes for the other, and every result reports its effective fold count alongside its nominal one.

4. The headline: testing the procedure, not the patterns

4.1 The reframing

The natural design is *find the best pattern in training, test it in the test window*. That yields **one out-of-sample test per fold**. Run 31.9M candidates through it and the best survivor across 8 folds is still very likely luck.

The stronger design inverts what is under test:

Across all folds, how often does a pattern selected in training actually win in testing?

This measures the **search procedure**, not any pattern. And it makes scan width an asset rather than a liability: the wider the search, the more precisely you measure how badly searching overfits.

4.2 What gets computed

For each fold, rank every candidate by its training statistic, take the top $N \in \{1, 10, 100\}$, and measure their performance in the untouched test window.

Metric	Question
Out-of-sample hit rate	Fraction of folds where the training-selected set beat zero
Degradation	Training statistic minus test statistic — how much was illusion
PBO	Probability of backtest overfitting (CSCV logit- λ)
Rank decay	Do training ranks predict test ranks at all?

4.3 Both outcomes are results

Outcome	Meaning
Hit rate $\approx$ 50%	Mass scanning does not work on this data. This is the most likely outcome and it is a genuine methodological finding — it generalises well beyond this dataset.
Hit rate $>$ 50%	The selection procedure carries information. Report the hit rate as headline, surviving patterns as secondary.

Crucially the bar here is **not** deflated by the number of cells scanned. There is exactly one procedure under test per configuration — a family of three, one per value of N . Scanning 31.9M candidates does not make the procedure test harder to pass; it makes it more precise.

5. The only viable formulation: cross-sectional persistence

Section 3.2 leaves exactly one estimator standing. Instead of "*is trading day 47 good?*" — which is undefined — the question must be:

Does trading day 47 rank stocks consistently — and does that ranking persist out of sample?

Measured as rank information coefficient: the Spearman correlation between the signal's cross-sectional ranking and the subsequent return ranking, with persistence tested as sign agreement across folds.

A persistent cross-sectional *ordering* is far harder to produce by chance than a persistent average, because chance does not usually order four thousand things the same way twice.

5.1 The unit of evidence is the date, not the stock-year

This is what the artifact in §3.2 was obscuring. **Each date yields one IC observation, however many stocks it ranks.** Pooling stocks buys precision *within* a date; it does not buy more dates. So the sample size is the number of independent dates — and that is why the 21,000 "stock-years" figure was always a mirage.

Measured 2026-08-18 on 656 stocks × 2,870 sessions:

Quantity	Value
IC observations	568
sd of IC	0.1190
SE(mean IC)	0.0050
MDE on mean IC	0.0140

(Worked example only — 21-session momentum against 5-session forward returns. The forward-return alignment in that probe was not audited, so the -0.0327 mean IC it produced is not reported as a finding.)

The important part is the floor: ~0.014. Real equity signal ICs typically run 0.02–0.05, so this estimator can see a genuine signal. That is more than can be said for any other formulation considered in this plan, and it is the reason Track S is worth building at all.

6. Track S1 — calendar patterns

Grid: 13 windows × 4 alignments × 202 indices and 4,200 stocks × 2 bases.

The null is not 50%, and it moves. The Indian market drifts upward, so the probability that a window closes higher rises with window length purely from drift. Measured by the predecessor: **0.461 at one day rising to 0.501 at ninety days.** A scan scored against a flat 50% would manufacture thousands of false discoveries at short windows and miss real ones at long. The null is measured by rotation, 1,000 rotations for final runs.

Leap-day cells are excluded. Near-duplicate cells — adjacent windows whose returns correlate above 0.90 — are grouped so that one effect is not counted thirteen times.

Turn-of-month and turn-of-year are removed from consideration, by owner decision Q43 and independently by arithmetic: **+129.70 bps gross against 136.06 bps of cost is –6.36 bps net.**

7. Track S2 — signal combinations

Owner decision, 2026-08-18: scan wide, explicitly to measure overfitting rather than in the expectation of finding an edge.

Seven families, roughly 190 base variants, crossed to depth 3 with thresholds. The realised combination count is computed at run time and sets the bar; a test greps for hard-coded literals so the count can never be stale.

Family	Variants	Mechanism
momentum	~40	underreaction to news
reversal	~30	liquidity provision is paid
volatility	~25	risk compensation / low-vol anomaly
volume	~25	attention and participation
liquidity	~20	illiquidity premium
seasonal	~30	flow calendar effects
institutional	~20	Track D deal signals as inputs

Every base signal requires a **written mechanism** before entering the pool. The scan is deliberately wide; that is not a licence to include things nobody can explain.

The prior is explicit and unflattering. V2 ran a strategy factory of exactly this shape for five weeks and promoted **zero** strategies. That is the starting expectation here, not a surprise to be explained away afterwards.

Track D's deal signals appear here as *inputs*. Track D remains a separate track with its own verdicts — this is reuse, not a merger.

8. The multiplicity bar

The required standard rises only with the **logarithm** of scan width, so a wide scan costs less than intuition suggests:

Corrected 2026-08-18 after three errors were found in the estimator, all pointing the anti-conservative way. See §8.1.

Width	Noise max t (normal)	Bar (normal)	Bar at df=20
5,000	3.87	4.84	5.93
3,146	3.75	4.68	5.73
1,000,000	4.98	6.23	8.60
31,893,556	5.63	7.04	11.27

A calendar cell scored on 21 yearly observations is $t(20)$, not normal, so the **df=20** column is the applicable one. For context, the predecessor's single best pattern sat at the 94th percentile of rotated noise — nowhere near any of these.

8.1 Three errors in the estimator, all anti-conservative

The bar published a day earlier was $|t| \geq 6.89$. It was wrong three ways, and every one made results *easier* to pass:

1. **Sidedness.** The estimator computed the expected maximum of *signed* normals while `Bar.clears()` compares `abs(t)`. Measured at N=171: `max(z)` is 2.693 but `max|z|` is 2.922. Every bar the module ever produced was low.
2. **Degrees of freedom.** Statistics are t-distributed, not normal, and 21 yearly observations give `t(20)`, whose tails are far fatter. Over 3,146 draws: normal 3.746 versus `t(20)` 4.599, **+23%**.
3. **Grid geometry.** Correlation between overlapping cells pulls the maximum *down* — measured 4.595 at $\rho=0$ falling to 3.071 at $\rho=0.7$ — partly cancelling effect 2. On a realistic simulated grid the truth was **4.151** against a dof-adjusted 4.60 and a normal 3.568.

No formula gets a specific grid right. Every Track S scan must therefore generate its own grid under its own null and measure the maximum directly (`multiplicity.simulated_max_null_t`). The table above is a planning guide, not the operative bar.

9. Costs, and a verdict that did not exist before

Every surviving pattern is costed before being reported, using the corrected schedule in `costs.yml`. The predecessor's headline seasonal finding was an artefact of a fee model wrong by 10.04 bps per round trip.

A new verdict is introduced: **SIGNIFICANT_BUT_UNPROFITABLE** — statistically real, economically negative. It is neither a pass nor a fail, and it was the true status of both of V2's surviving seasonal effects.

10. Compute, measured rather than asserted

The predecessor's "~3 weeks" for a full rescan was an assertion; no single cell was ever timed. The dual-fold design multiplies the work again.

Before any full run, a benchmark on 1/1000th of the grid must produce a measured projection. If that projection exceeds the 21-day budget, the grid is cut and the cut is recorded as a decision rather than absorbed silently.

10a. How Track S connects to Track D

Written 2026-08-18 after a check found the two tracks had never been wired together. Three gaps, each of which would have surfaced only once scan code ran. Full record: decision 0023.

10a.1 Trial families — who pays for which search

`research.yml` said the trial counter applied to **everything**. `scan.yml` said nothing. Read literally, running Track S once would have done this:

Counter state	Track D bar
today (171 trials)	3.71
after a 5,000-combination signal scan	4.92
after a 1M-cell calendar scan	6.42
after the full 31.9M rescan	7.28

exp_001's $t = -3.93$ would have **retroactively failed**, and no deal study could ever have passed again. **Track S would have destroyed Track D as collateral damage**, and nobody would have found out until the day a scan ran.

The counter is now hierarchical (`configs/trials.yml`). Four families:

Family	Charges	dof	Carried
TRACK_D_DEALS	deal event studies	246	171
TRACK_S_CALENDAR	calendar cells	20	31,893,556
TRACK_S_SIGNALS	signal combinations	~500	0
TRACK_S_PROCEDURE	the procedure test	—	fixed at 3

TRACK_S_CALENDAR carries the predecessor's completed 31.9M-cell scan. **That space is not virgin**, and a rebuild does not get to look at it as though for the first time.

The procedure exemption. TRACK_S_PROCEDURE stays at a family size of 3 — one per value of N — no matter how wide the scan. Exactly one procedure is under test per configuration, and the 31.9M cells are the **instrument** that measures it, not competing hypotheses. This is precisely what makes §4's "scan wide to measure overfitting" a legitimate design rather than a loophole, and it holds **only** while the reported claim is about the procedure. The moment a specific surviving pattern is reported, that claim pays full width in TRACK_S_CALENDAR.

A project-level claim faces everything. A within-family bar answers "is this the best of the calendar cells". A claim that *the project* found something is a selection across all four families and faces the summed bar. Both are always reported — publishing only the friendlier one is exactly what the predecessor did when it exempted its own champion.

What stops family-splitting being a loophole is declaration order, not family size. Families are declared before the search and are immutable afterwards, and a result may never be moved to a smaller family once seen.

10a.2 The Track S partition

`split.yml` partitions by ISIN, which is meaningless for a calendar cell — a cell spans every stock at once. So Track S had **no exploration/confirmation regime at all** while Track D sat behind a guard that raises.

Split	Role	Boundary
time	mandatory	explore $\leq$ 2015-12-31, confirm $\geq$ 2016-01-01, 21-session embargo
index	corroborating only	40 / 60 by hash of index name

The time split is mandatory because it is **the only partition that tests persistence**, which is what a pattern claim asserts. The index split is explicitly weak and labelled as such: the 202 indices overlap heavily, since NIFTY 50 constituents sit inside NIFTY 100, NIFTY 500 and most sector and thematic indices, so an "explore" index and a "confirm" index can share most of their members.

A cell must clear its bar under **both**, and observation floors apply per half.

10a.3 The design gate now covers scans

`StudyKind` gained `scan`, and five blocking confounds were added for it — `multiple_testing_declared`, `null_is_measured_not_assumed`, `fold_independence`, `bid_ask_bounce`, `prior_search_of_this_space`. Before then the gate applied to Track D and left the half of the project with far worse multiple-testing exposure with no required controls at all.

A scan must declare its family, its **nominal** fold count and its **effective** fold count before registration. A single-fold scan is refused: one fold is an in-sample fit, which is what the 31.9M atlas was.

10a.4 A fourth gap, found inside the fix

`trials.yml` declared the counters *monotonic* and *never reset* — and nothing incremented them. `charge()` was a pure function and the project counter summed static YAML, so a 31.9M-cell scan could have run without moving anything.

That is `exp_001`'s `trials_before` — computed, stored, printed once, never read — **rebuilt one level up, inside the file whose subject is that exact failure**. Fixed by migration `0002_trial_families`: an append-only `family_charge` ledger whose triggers refuse UPDATE, DELETE, and any total that would decrease.

11. What has to be built

Module	Purpose	Status
<code>src/scan/folds.py</code>	Anchored walk-forward + CPCV generation, purge/embargo	not built
<code>src/scan/nulls.py</code>	Measured rotation null	not built
<code>src/scan/calendar.py</code>	S1 cell enumeration and scoring	not built
<code>src/scan/signals.py</code>	S2 combination enumeration	not built
<code>src/scan/procedure.py</code>	The headline: hit rate, degradation, PBO, rank decay	not built
<code>src/scan/atlas.py</code>	Chunked resumable execution, checkpointing	not built
migration 0002	<code>scan_cell</code> , <code>scan_fold_result</code> , <code>procedure_result</code>	not built

Reused unchanged from Track D: `multiplicity.py`, `hashing.py`, `migrate.py`, `paths.py`, the experiment registry and the decision-record discipline.

Not reused: `split.py` and `power.py`. Both are event-study machinery — the name-based partition is meaningless for a calendar cell, and the monthly cohort collapse does not apply to annual observations. Track S needs its own equivalents, which is most of the build above.

12. Honest expectation

Stated in advance so it cannot be adjusted afterwards:

- **The procedure test most likely returns a hit rate near 50%**, meaning mass scanning does not work on this data. That is the primary expected finding and it is worth having.
- **Very few or no individual patterns will clear the simulated bar**, which on a realistic grid lands near $|t| \geq 5.2$ and rises steeply with width. The predecessor's best cleared nothing close.
- **The calendar track is the weaker of the two.** It has been run before and returned nothing, and its per-stock form is arithmetically dead.
- **The signal track has better arithmetic** — signals fire many times a year, so walk-forward works natively — but its prior is a factory that promoted zero.

The value of Track S does not depend on finding an edge. It depends on producing a defensible measurement of whether searching for one works at all, on twenty-one years of real data, with the costs modelled correctly.

That measurement does not currently exist anywhere in this project, and it is the thing the 31.9M cells are actually good for.